

Integration 1 HW

Review

1.
 - i. False
 - ii. True
 - iii. True

2.
$$x^2 + (k - 1)x + 4 = 0$$

If the equation has at least one real root, the discriminant is greater than zero. Then

$$\begin{aligned}b^2 - 4ac &\geq 0 \\(k - 1)^2 - 4(1)(4) &\geq 0 \\k^2 - 2k - 15 &\geq 0 \\(k - 7)(k + 5) &\geq 0 \\k &= 7 \text{ or } k = -5\end{aligned}$$

And since the coefficient of x^2 is greater than zero, $-5 \leq k \leq 7$.

3. The derivative of $(2x - 1)(x^2 - 3)$ is $\frac{d}{dx}(2x^3 - x^2 - 6x + 3) = 6x^2 - 2x - 6$ **(b)**

4.
$$y = x^3 + 2x^2 - 3x + 1$$
$$\frac{d^2y}{dx^2} = 6x + 4$$

When $x = -2$, $6(-2) + 4 = -8$

5.
$$2 \cos \theta - \sin \theta = 0$$
$$\frac{\sin \theta}{2 \cos \theta} = 1$$
$$\tan \theta = 2$$
$$\theta = 63^\circ, 243^\circ$$

Consolidation

1.
$$\int \left(2\sqrt{x} - \frac{3}{\sqrt{x}} \right) = \int (2x^{\frac{1}{2}} - 3x^{-\frac{1}{2}}) = \frac{4}{3}x^{\frac{3}{2}} + 6x^{\frac{1}{2}} + c$$

2.
$$f(x) = 2x^3 - x^2 + 3x + c$$
$$\therefore f(1) = 2 - 1 + 3 + c = 2$$
$$\therefore c = -2$$

So $y = 2x^3 - x^2 + 3x - 2$

3. $f'(x) = x^{-2} - x^{\frac{3}{2}}$ and $f(1) = 2$.

Therefore $y = -\frac{1}{x} - \frac{2}{5}x^{\frac{5}{2}} + c$.

When $x = 1$, $y = 2$. So

$$2 = -1 - \frac{2}{5} + c$$

$$c = \frac{17}{5}$$

Thus $y = -\frac{1}{x} - \frac{2}{5}x^{\frac{5}{2}} + \frac{17}{5}$.

4. i. $y = x^3 - 4x + c$

ii. $y = x^3 - 4x - 1$

iii. When $x = 1$,

$$y = 1^3 - 4(1) - 1$$

$$= 1 - 4 - 1$$

$$= -4$$

So the curve passes through $(1, -4)$

5. $h = 25t - 5t^2 + c$.

At $t = 0$, $h = 30$, so $c = 30$.

Then when $h = 0$,

$$-5t^2 + 25t + 30 = 0$$

$$(30 - 5t)(t + 1) = 0$$

$$t = 6 \text{ or } t = -1$$

But since $t \geq 0$, $t = 6$.

6. At maximum point, $\frac{dy}{dx} = 0$

i. $3x^2 - 2x + k = 0$

$$12 + 4 + k = 0$$

$$k = -16$$

ii. $y = x^3 - x^2 - 16x + c$.

At $x = 1$:

$$3 = 1^3 - 1^2 - 16(1) + c$$

$$c = 19$$

So the equation of the curve is $y = x^3 - x^2 - 16x + 19$

Exam Questions

1. a. $\int (x^3 - 6x) dx = \frac{1}{4}x^4 - 3x^2 + c$

b. $\int \left(\frac{4}{x^2} - 1 \right) dx = \int (4x^{-2} - 1) = -\frac{4}{x} - x$

2.
$$\frac{dy}{dx} = 2x - 4$$
$$\therefore y = x^2 - 4x + c$$

Since the curve goes through point $(2, 7)$,

$$7 = 2^2 - 4(2) + c$$
$$c = 11$$

So the equation of the curve is $y = x^2 - 4x + 11$

3. $P \Rightarrow Q$ (the first statement)